Constraints on Higgs-charm couplings at CMS

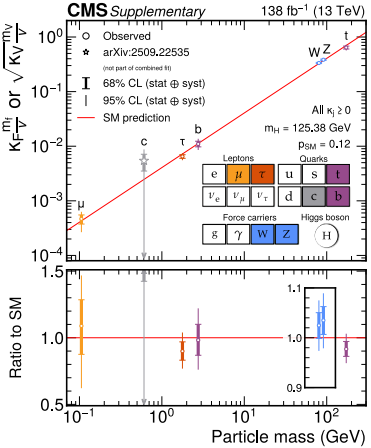
Chirayu Gupta
on behalf of the CMS Collaboration

IIHE, Vrije Universiteit Brussel (VUB)

Higgs 2026, Louvain-la-Neuve, 28 October 2026

Why the Higgs-charm coupling?

- Since the 2012 discovery: couplings to t, b, τ and μ measured, all SM-like
- **2nd-generation quark Yukawas are the open frontier**; charm is the most accessible
  - $\mathcal{B}(\mathrm{H} \to \mathrm{c\bar{c}}) \approx 2.9\%$, about $20\times$ below $\mathrm{H} \to \mathrm{b\bar{b}}$
  - huge QCD backgrounds; c jets are hard to tell from b and light jets
- $\kappa_\mathrm{c} = y_\mathrm{c}/y_\mathrm{c}^\mathrm{SM}$ (SM: $\kappa_\mathrm{c} = 1$)
- Flavour-compatible BSM (e.g. spontaneous flavour violation) allows $|\kappa_\mathrm{c}| \sim 3$: current sensitivity is entering this region
- Does the Higgs mechanism also give the 2nd-generation quarks their mass?



CMS, Rep. Prog. Phys. 89 (2026) 087801 (supplementary)

How CMS probes κ_c

Direct $H \to c\bar{c}$ (rate $\propto \kappa_c^2/\Gamma_H$) slides 4–6

VH($H \to c\bar{c}$)	PRL 131 (2023) 061801
$t\bar{t}H(H \to c\bar{c})$ + VH comb.	PRL 136 (2026) 011801
Boosted $H \to c\bar{c}$ (mostly ggH)	PRL 131 (2023) 041801

best: $|\kappa_c| < 3.5$ **(2.7 exp.)**

H + charm production (cg $\to$ cH) slides 7–8

cH($H \to \gamma\gamma$)	JHEP 11 (2025) 060
cH($H \to WW$) + comb.	JHEP 08 (2026) 051

best: $|\kappa_c| < 47$ **(51 exp.)**

Indirect: H rates and width slide 9

On-shell $H \to 4\ell$ (γH paper)	PRD 112 (2025) 112001
Off-shell + on-shell $H \to 4\ell$ **NEW**	arXiv:2607.23352

$\kappa_c \in [-3.3, 2.8]$ **to** $[-8.6, 7.5]$ (by assumptions)

Rare exclusive decays slide 9

$H \to J/\psi\gamma$, $\psi(2S)\gamma$ PLB 865 (2025) 139462

sensitive to the *sign* of κ_c; see J. Bierkens' talk

$-166 < \kappa_c < 208$

$\mu = \sigma\mathcal{B}/(\sigma\mathcal{B})_{SM}$, Γ_H = total Higgs width, limits at 95% CL: obs. (exp.)

All κ_c results use **Run 2** data (13 TeV, 138 fb^{-1}); Run 3 has delivered the tools (slide 11); **NEW** = public in 2026

Direct $\mathrm{H} \to \mathrm{c\bar{c}}$: VH and boosted H

- **VH**: leptonic W/Z; resolved + merged jets, ParticleNet c tagging
- VZ($\mathrm{Z} \to \mathrm{c\bar{c}}$) seen at 5.7σ

VH($\mathrm{H} \to \mathrm{c\bar{c}}$)

95% CL obs. (exp.)

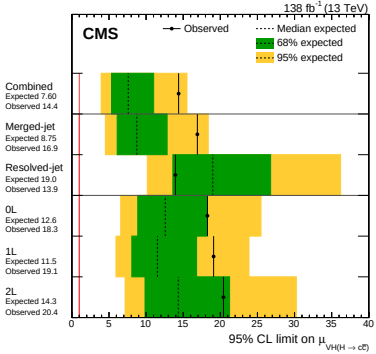
$\mu < 14\,(7.6)$

$1.1 < |\kappa_\mathrm{c}| < 5.5 \;\; (< 3.4)$

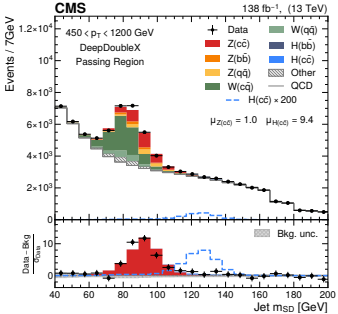
obs. excludes $\kappa_\mathrm{c} = 1$: upward fluctuation, $\hat{\mu} = 7.7$ → backup

Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$

$\mu < 47\,(39)$ ($p_\mathrm{T} > 450$ GeV, mostly ggH)



VH: μ limits per channel



Boosted: jet mass, passing region (peak: $\mathrm{Z} \to \mathrm{c\bar{c}}$)

VH carries the Run 2 sensitivity; boosted ggH adds an independent high-p_T handle

$t\bar{t}H(H \to c\bar{c})$: a new channel

- 0, 1, 2-lepton $t\bar{t}$ decays; many jets of all flavours
- Jets in b/c/light **2D tagging categories** (ParticleNet); transformer event classifier (ParT)
- **Simultaneous** fit of $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z$; $t\bar{t}+c\bar{c}/b\bar{b}$ from control regions
- Check: $t\bar{t}Z(Z \to c\bar{c})$ $\mu = 1.0 \pm 0.8$ → backup

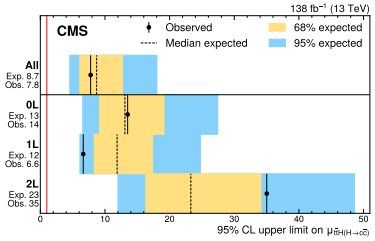
$t\bar{t}H(H \to c\bar{c})$

95% CL obs. (exp.)

$\mu < 7.8\,(8.7)$

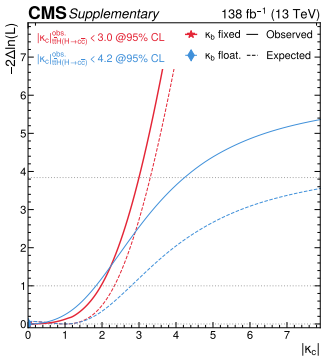
$|\kappa_c| < 3.0\,(3.3)$ $(\kappa_b = 1)$

$|\kappa_c| < 4.2$ obs. (κ_b fitted)



$\mu_{t\bar{t}H(c\bar{c})}$ limits per channel

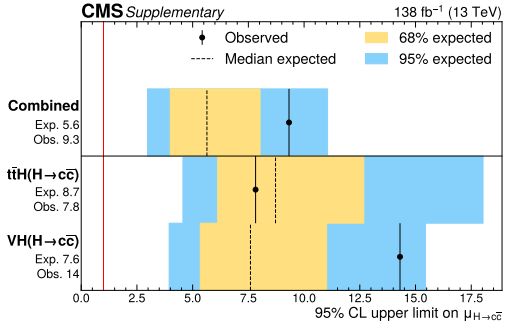
PRL 136 (2026) 011801, CMS-HIG-24-018

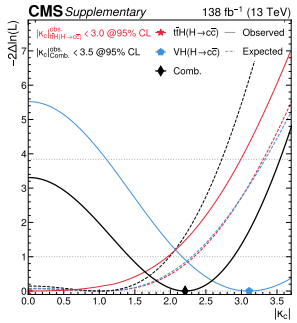


$-2\Delta \ln L$ vs. $|\kappa_c|$, κ_b fixed or fitted from $t\bar{t}H(b\bar{b})$

A second direct channel, with expected sensitivity comparable to VH

Combination of $\mathrm{t\bar{t}H}$ and VH: most stringent *direct* constraint on κ_c





PRL 136 (2026) 011801, CMS-HIG-24-018

$\mathrm{t\bar{t}H}$ + VH

95% CL obs. (exp.)

$\mu_{\mathrm{H}\to\mathrm{c\bar{c}}} < 9.3$ (5.6)

$|\kappa_c| < 3.5$ (2.7)

SM production, $\kappa_b = 1$

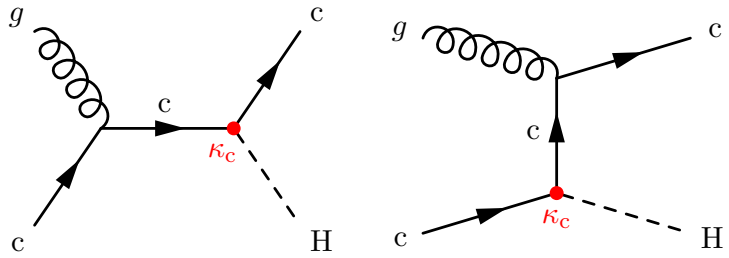
- Expected μ limit 26% better than VH alone (36% better than $\mathrm{t\bar{t}H}$ alone)
- $\mathrm{t\bar{t}H}$ alone (3.0) beats the combination (3.5) on observed because VH fluctuated up ($\hat{\mu} = 7.7$)

Best expected sensitivity to date, $|\kappa_c| < 2.7$: below the BSM-motivated $|\kappa_c| \sim 3$

H + charm production: κ_c in production

JHEP 11 (2025) 060, CMS-HIG-23-010

- κ_c enters the **production** vertex, independent of the $H \to c\bar{c}$ decay: clean H decay + a c-tagged jet
- $\sigma_{cH} = 90\,\text{fb}$: $\sigma\mathcal{B} \approx 0.2\,\text{fb}$ ($\gamma\gamma$), $\approx 20\,\text{fb}$ (WW)
- Challenges: κ_c-**independent H+c** from g$\to c\bar{c}$ splitting in ggH (same final state); soft, often forward c jets; flavour-scheme modelling ($\to$ T. Bevilacqua, next talk)
- **"Flat direction"**: κ_c is varied while all other Higgs rates stay at their SM values, so only cH production changes



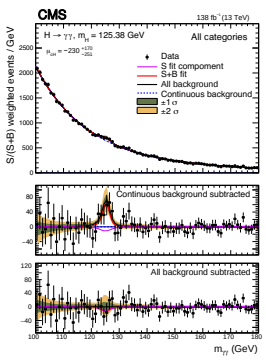
LO diagrams for cH production; κ_c enters at the marked vertices

cH production: H→ $\gamma\gamma$, H→WW and combination

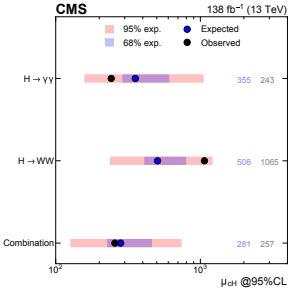
- **H→ $\gamma\gamma$**: BDTs vs. ggH and vs. $\gamma\gamma$+jets (no c-tag inputs), 27 categories, fit to $m_{\gamma\gamma}$
  $\mu_{\mathrm{cH}} < 243\,(355)$, $|\kappa_{\mathrm{c}}| < 38\,(73)$
- **H→WW→e$\nu\mu\nu$ + c jet**: BDTs vs. non-H and other-H backgrounds, signal and control regions
  $\mu_{\mathrm{cH}} < 1065\,(506)$, $|\kappa_{\mathrm{c}}| < 211\,(95)$
- H+c background fixed to SM; floating it → backup

$\gamma\gamma$ + WW combination (95% CL)

$\mu_{\mathrm{cH}} < 257\,(281)$, $|\kappa_{\mathrm{c}}| < 47\,(51)$



$\gamma\gamma$: $m_{\gamma\gamma}$, all categories, weighted by S/(S+B)

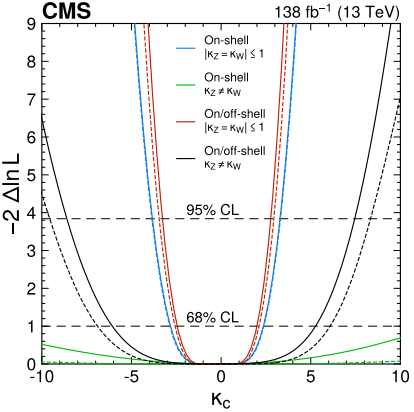


μ_{cH} limits: $\gamma\gamma$, WW, combined

$|\kappa_{\mathrm{c}}| \lesssim 50$, not $\sim$3: $\sigma\mathcal{B}$ is tiny and ggH H+c mimics the signal; statistics-limited

Indirect constraints on κ_c

- Large κ_c adds $c\bar{c} \to H$ production and increases Γ_H: on-shell $H \to 4\ell$ rates constrain it
- **On-shell $H \to 4\ell$** (γH paper), $|\kappa_Z = \kappa_W| \leq 1$: $\kappa_c \in [-3.7, 3.2]$ ($[-3.8, 3.3]$)
- NEW **Off-shell + on-shell $H \to 4\ell$**: the off-shell rate breaks the κ_V–Γ_H degeneracy
  - $|\kappa_V| \leq 1$: $\kappa_c \in [-3.3, 2.8]$ ($[-3.5, 2.9]$)
  - κ_Z, κ_W free: $\kappa_c \in [-8.6, 7.5]$ ($[-9.6, 8.3]$)
  - width measurement: F. Errico's talk
- $H \to J/\psi\gamma$: $-166 < \kappa_c < 208$ (J. Bierkens' talk)



κ_c scans, solid = observed, dashed = expected

With $|\kappa_V| \leq 1$, $H \to 4\ell$ matches the direct searches, under different assumptions → backup

Where we stand: CMS constraints on κ_c

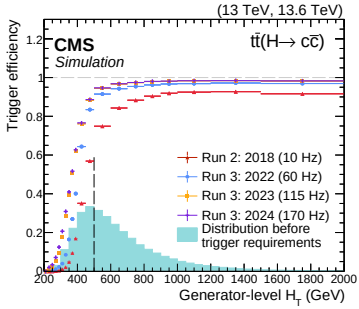
Probe	Reference	κ_c at 95% CL: obs. (exp.)	Main assumptions
$t\bar{t}H$ + VH($H \to c\bar{c}$)	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.5$ (2.7)	$\kappa_b = 1$, SM prod., no BSM decays
$t\bar{t}H$($H \to c\bar{c}$)	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.0$ (3.3)	same
VH($H \to c\bar{c}$)	PRL 131 (2023) 061801	$1.1 < \lvert\kappa_c\rvert < 5.5$ (3.4)	same
On+off-shell H$\to 4\ell$	arXiv:2607.23352	$[-3.3, 2.8]$ ($[-3.5, 2.9]$)	$\lvert\kappa_Z = \kappa_W\rvert \leq 1$, $\kappa_{b,t} = 1$, light κ_q prof.
On-shell H$\to 4\ell$	PRD 112 (2025) 112001	$[-3.7, 3.2]$ ($[-3.8, 3.3]$)	same
On+off-shell H$\to 4\ell$	arXiv:2607.23352	$[-8.6, 7.5]$ ($[-9.6, 8.3]$)	κ_Z, κ_W free
cH ($\gamma\gamma$ + WW)	JHEP 08 (2026) 051	$\lvert\kappa_c\rvert < 47$ (51)	flat direction, from μ_{cH} limit
H$\to$J/$\psi\gamma$	PLB 865 (2025) 139462	$[-166, 208]$	$\kappa_\gamma = 1$

The rows use different coupling models and are not directly comparable.

- $t\bar{t}H$ alone beats the combination on observed because VH fluctuated up; the combination has the best *expected* sensitivity (2.7)
- Direct searches and off-shell + on-shell H$\to 4\ell$ both reach $\lvert\kappa_c\rvert \sim 3$, where flavour-compatible BSM scenarios live

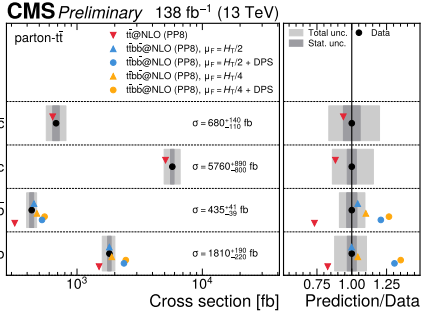
Toward Run 3: better triggers, measured backgrounds

ML b/c tagging at the HLT



arXiv:2608.18331: $t\bar{t}H(H \to c\bar{c})$ trigger efficiency, Run 2 (2018) vs. Run 3 (2022–24)

Main $t\bar{t}H(H \to c\bar{c})$ background measured



CMS-PAS-TOP-26-003 **NEW**: $\sigma_{t\bar{t}c\bar{c}} = 680^{+130}_{-100}\,(\mathrm{syst}) \pm 40\,(\mathrm{stat})$ fb

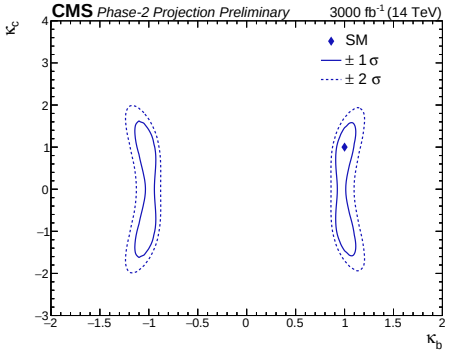
- 13.6 TeV; new HLT taggers (see J. Daniel's talk), parking and scouting for hadronic final states → backup

Run 3 H→ $c\bar{c}$ searches start with higher trigger efficiency and a measured $t\bar{t}$+$c\bar{c}$ background

Prospects at the HL-LHC

- ATLAS + CMS input to the European Strategy update
  (arXiv:2504.00672, CMS-HIG-25-002), $3\,\mathrm{ab}^{-1}$ per experiment:
  - VH($\mathrm{H} \to \mathrm{c\bar{c}}$): $\approx 1.6\sigma$ expected
  - **expected** $|\kappa_\mathrm{c}| < 1.5$ at 95% CL, 30% better than the previous direct-search projection
- Expected $\mu_\mathrm{H} \to \mathrm{c\bar{c}}$ limit: ∼37 (2016 data, JHEP 03 (2020) 131) → 5.6 (Run 2 VH+$\mathrm{t\bar{t}H}$), much faster than luminosity alone
- Next gains: Run 3 data, ML tagging, detector upgrades; theory modelling and tagger calibration become limiting

Goal for the HL-LHC: κ_c at $\mathcal{O}(1)$, testing whether the Higgs gives the charm quark its mass



Previous CMS-only projection (HIG-21-008 supplementary): expected $\pm 1\sigma/\pm 2\sigma$ (κ_b, κ_c) contours, 3000 fb^{-1}, 14 TeV, extrapolated from Run 2 VH($\mathrm{H} \to \mathrm{c\bar{c}}$)

Summary

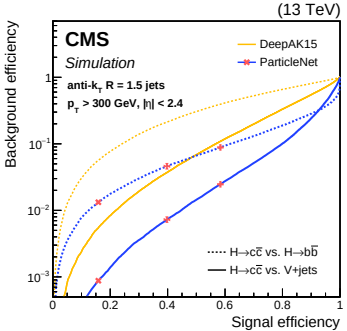
- The charm Yukawa is the most accessible 2nd-generation quark coupling; CMS probes it in decay, in production and indirectly
- $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ + $\mathrm{VH}(\mathrm{H} \to \mathrm{c\bar{c}})$: $|\kappa_\mathrm{c}| < 3.5$ **(2.7 exp.)**, the most stringent direct constraint and the best expected sensitivity
- H+c production: $|\kappa_\mathrm{c}| < 47$ (51 exp.), independent of the $\mathrm{H} \to \mathrm{c\bar{c}}$ decay
- Off-shell + on-shell $\mathrm{H} \to 4\ell$: $\kappa_\mathrm{c} \in [-3.3, 2.8]$ with $|\kappa_V| \leq 1$ (about ± 8 without)
- All κ_c results are Run 2; Run 3 brings higher trigger efficiency and a measured $\mathrm{t\bar{t}+c\bar{c}}$ background
- **Does the Higgs give charm its mass?** Today $|\kappa_\mathrm{c}| \lesssim 3$–$3.5$, entering BSM-motivated territory; the HL-LHC is expected to reach $|\kappa_\mathrm{c}| < 1.5$

Thank you!

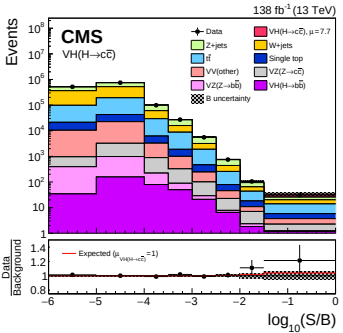
Backup

VH($\mathrm{H} \to \mathrm{c\bar{c}}$): event categories and sensitivity

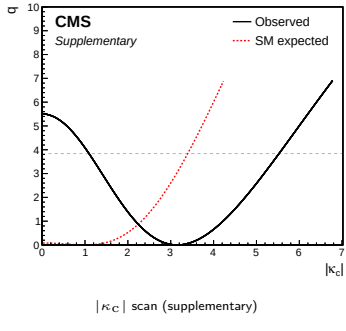
PRL 131 (2023) 061801



ParticleNet merged-jet $\mathrm{H} \to \mathrm{c\bar{c}}$ tagging

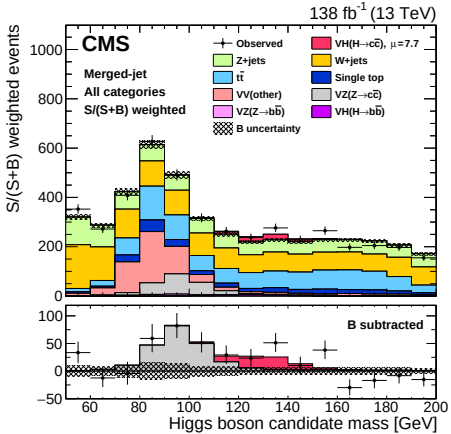


Events vs. $\log_{10}(S/B)$

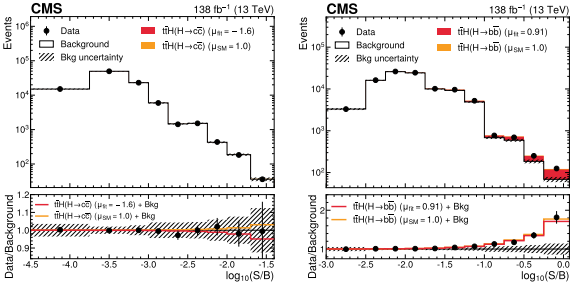


Best fit $\mu = 7.7^{+3.8}_{-3.5}$; merged $\mu = 8.7^{+4.6}_{-4.0}$, resolved $\mu = -9.5 \pm 9.6$. $\sigma\mathcal{B} < 0.94$ (0.50) pb.
μ limits with CLs; κ_c intervals from profile-likelihood scans.

VH($\mathrm{H} \to \mathrm{c\bar{c}}$) merged jets and $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ yields



PRL 131 (2023) 061801: VH merged-jet $m(\mathrm{H_{cand}})$, events weighted by S/(S+B)

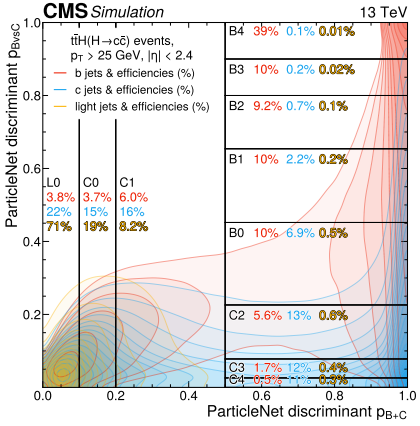


PRL 136 (2026) 011801: event yields vs. $\log_{10}(S/B)$; the $\mathrm{t\bar{t}H}(\mathrm{c\bar{c}})$ panel shows the published best fit $\hat{\mu} = -1.6 \pm 4.5$

$t\bar{t}H(H \to c\bar{c})$: flavour tagging and by-products

PRL 136 (2026) 011801

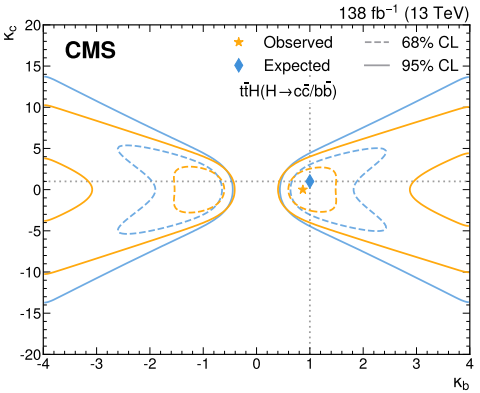
- Each jet gets ParticleNet scores; the (b+c vs. light, b vs. c) plane is split into 2D categories
- ParT event classifier separates $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z$, $t\bar{t}$+HF
- $t\bar{t}Z(Z \to c\bar{c})$ as a standard candle: $\mu = 1.0 \pm 0.8$
- By-product: $t\bar{t}H(H \to b\bar{b})$ $\mu = 0.91^{+0.26}_{-0.22}$, 4.4σ (4.5 exp.)



b, c, light jets in the ParticleNet (b+c vs. light, b vs. c) plane and the 2D tagging categories

$t\bar{t}H(H \to c\bar{c}/H \to b\bar{b})$: (κ_b, κ_c) plane

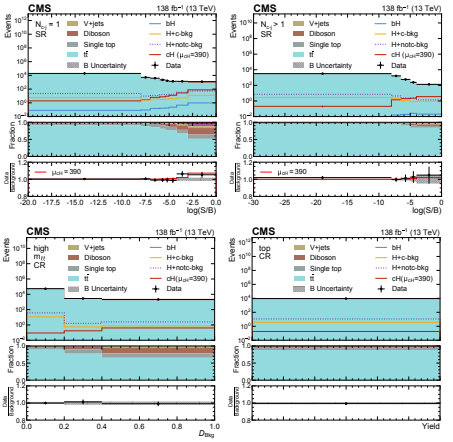
PRL 136 (2026) 011801



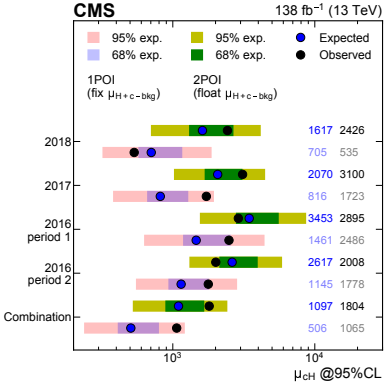
orange: observed, blue: expected; dashed 68%, solid 95% CL

- Simultaneous $t\bar{t}H(H \to b\bar{b})$ + $t\bar{t}H(H \to c\bar{c})$ fit constrains both Yukawas
- Signal strengths: $t\bar{t}H(b\bar{b})$ $0.91^{+0.26}_{-0.22}$; $t\bar{t}H(c\bar{c})$ -1.6 ± 4.5; $t\bar{t}Z(b\bar{b})$ $1.47^{+0.45}_{-0.41}$; $t\bar{t}Z(c\bar{c})$ $1.02^{+0.79}_{-0.84}$

cH(H→WW): post-fit distributions and per-period limits



1-POI post-fit distributions of the BDT discriminants



μ_{cH} limits per data-taking period, 1 and 2 POI

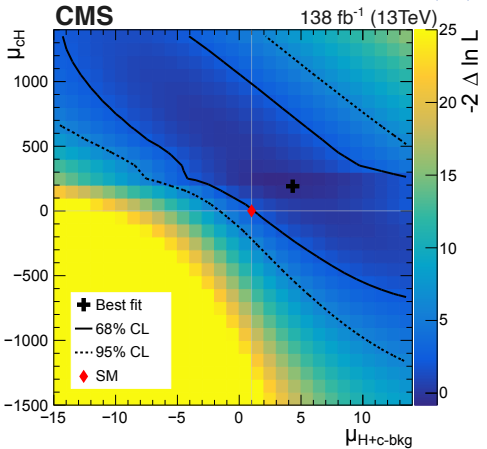
JHEP 08 (2026) 051

Best fit (1 POI): $\mu_{\mathrm{cH}} = 390^{+380}_{-270}$, compatible with SM within 1.5σ; limited by data statistics, then flavour-scheme and ggH heavy-flavour modelling.

cH(H→WW): floating the H+c background

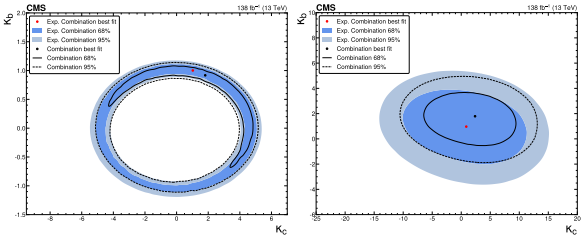
- 1 POI: only μ_{cH} floats; the κ_{c}-independent H+c background (from ggH etc.) is fixed to its SM prediction
- 2 POI: H+c background also floats; correlation with cH: -87%
- 2-POI limit: $\mu_{\mathrm{cH}} < 1804\,(1097)$
- The combination with $\gamma\gamma$ uses the 1-POI fit

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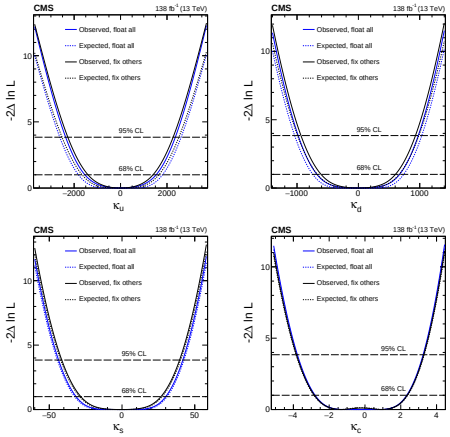


WW: μ_{cH} vs. $\mu_{\mathrm{H+c\ bkg}}$

Indirect: Higgs p_{T} spectrum and on-shell H$\to 4\ell$



JHEP 04 (2026) 093: (κ_{b}, κ_{c}) from the Run 2 $p_{\mathrm{T}}^{\mathrm{H}}$ combination ($\gamma\gamma$, ZZ, resolved $\tau\tau$); left $\mathcal{B}$ coupling-dependent, right $\mathcal{B}$ unconstrained; 2D contours only

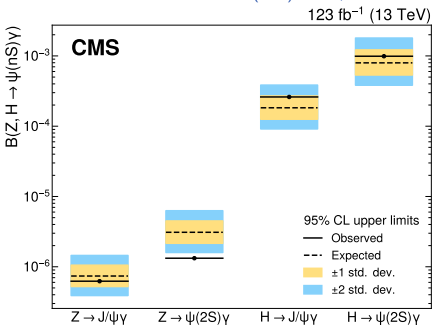


PRD 112 (2025) 112001 (γH + on-shell H$\to 4\ell$): light-quark κ scans

Rare exclusive decays: $H \to J/\psi\gamma$, $\psi(2S)\gamma$

- Direct ($\propto \kappa_c$) and indirect ($H \to \gamma\gamma^* \to \gamma\, J/\psi$) amplitudes interfere $\Rightarrow$ sign sensitivity
- $123\,\mathrm{fb}^{-1}$: $\mathcal{B}(H \to J/\psi\gamma) < 2.6 \times 10^{-4}$, $\mathcal{B}(H \to \psi(2S)\gamma) < 9.9 \times 10^{-4}$
- $\kappa_c/\kappa_\gamma \in (-157, 199)$; with $\kappa_\gamma = 1$: $\kappa_c \in (-166, 208)$

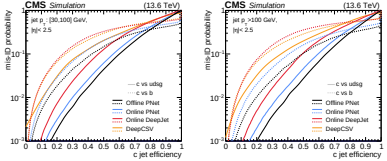
PLB 865 (2025) 139462, CMS-SMP-22-012



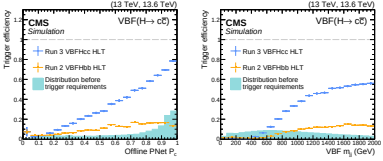
95% CL upper limits on $\mathcal{B}(Z, H \to \psi(nS)\gamma)$, obs. and exp.

Run 3 HLT flavour tagging

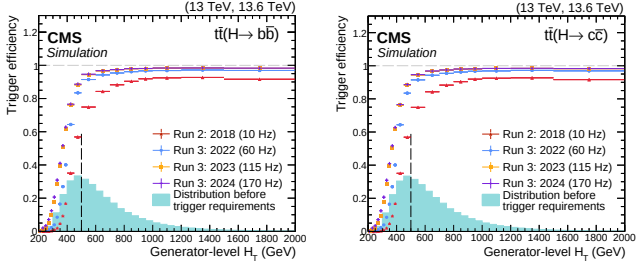
arXiv:2608.18331, CMS-BTV-25-002



HLT tagger misidentification vs. efficiency



VBF H→ $c\bar{c}$ trigger efficiency



$t\bar{t}H(H \rightarrow b\bar{b})$ (left) and $t\bar{t}H(H \rightarrow c\bar{c})$ (right) trigger efficiency, Run 2 vs. Run 3

References

- VH($\mathrm{H} \to \mathrm{c\bar{c}}$): CMS, PRL 131 (2023) 061801, arXiv:2205.05550 (HIG-21-008)
- $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ + VH combination: CMS, PRL 136 (2026) 011801, arXiv:2509.22535 (HIG-24-018)
- Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$: CMS, PRL 131 (2023) 041801, arXiv:2211.14181 (HIG-21-012)
- VH($\mathrm{H} \to \mathrm{c\bar{c}}$) with 2016 data: CMS, JHEP 03 (2020) 131 (HIG-18-031)
- cH(H$\to \gamma\gamma$): CMS, JHEP 11 (2025) 060, arXiv:2503.08797 (HIG-23-010)
- cH(H$\to$WW) and combination: CMS, JHEP 08 (2026) 051, arXiv:2508.14988 (HIG-24-009)
- Differential combination Run 2 (backup): CMS, JHEP 04 (2026) 093, arXiv:2504.13081 (HIG-23-013)
- γH + H$\to 4\ell$ light Yukawas: CMS, PRD 112 (2025) 112001, arXiv:2502.05665 (HIG-23-011)
- Off-shell + on-shell H$\to 4\ell$: CMS, arXiv:2607.23352, submitted to PRD (HIG-25-003)
- H$\to$J/$\psi\gamma$: CMS, PLB 865 (2025) 139462, arXiv:2411.15000 (SMP-22-012)
- HLT flavour tagging: CMS, arXiv:2608.18331 (BTV-25-002)
- $\mathrm{t\bar{t}}$+b/c cross sections: CMS-PAS-TOP-26-003
- HL-LHC: ATLAS and CMS, arXiv:2504.00672 (HIG-25-002)
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- Spontaneous flavour violation (motivation): arXiv:2410.05236